

```
mysql> show databases;
```

Database
cg_mysql
information_schema
mysql
performance_schema
praveendb
sys

```
6 rows in set (0.01 sec)
```

```
mysql> use cg_mysql;
```

```
Database changed
```

```
mysql> show tables;
```

Tables_in_cg_mysql
company_staff
employees

```
2 rows in set (0.01 sec)
```

```
mysql> /* Constraonts */
```

```
-> /* constraints are rules applied to a table columns in MYSQL to ensure that only valid and accurate data is stored in the database. */
```

```
-> /* in simple words:*/
```

```
-> /* constraints= rules that control the data entered into a table.*/
```

```
-> /* why do we use constraints? */
```

```
-> /* constraints are used to
```

```
-> 1. maintain data integrity prevent invalid data entry.
```

```
-> avoid duplicate records ensures relationships between tables, make the database reliable and consistent. */
```

```
-> ^C
```

```
mysql> /* integrity constraints
```

```
-> 1.not null
```

```
-> 2.unique
```

```
-> 3.primary key
```

```
-> 4. composite key
```

```
-> 5. foreign key
```

```
-> 6.default
```

```
-> 7. check */
```

```
-> ^C
```

```
mysql> /* not null: the not null constriants ensures that a column cannot contain Null (empty)values.every row must have a value for that column.
```

```
/*> not null: doesn't allow null values.
```

```
/*> unique: The UNIQUE key constrianr ensures that all values in a column are different.duplicate values are not allowed.
```

```
/*> Primary key: Combination of unique +not null */
```

```
mysql> /* Composite key: If we applied more than once column as primary key then we
should go for compsite key constraint this contain is applicable only for table
level not at column-level */
```

```
mysql> /* default key: the default constraint automatically assigns a predefined
value if no value is provided during insertion.*/
```

```
-> ^C
```

```
mysql> /* Check: The check constriant ensure that value satisfy a specified
condition before insertion or update. */
```

```
mysql> /* Foreign key: A foreign key creates a relationship between two tables. it
ensures that value in the child table must already exists in the parent table.*/
```

```
mysql> /* Referential actions
```

```
/*> referential actions define what happens to child table data when parent
table data is updated or deleted'
```

```
/*> (ON DELETE CASCADE, ON UPDATE CASCADE, ON DELETE SET NULL,ON UPDATE SET
NULL) */
```

```
mysql> ceeate table products(
```

```
-> product_id int primary key,
```

```
-> product_name varchar(100),
```

```
-> price decimal(10,2)
```

```
-> );
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'ceeate
table products(
```

```
product_id int primary key,
```

```
product_name varchar(100),
```

```
pr' at line 1
```

```
mysql> create table products(
```

```
-> product_id int primary key,
```

```
-> product_name varchar(100),
```

```
-> price decimal(10,2)
```

```
-> );
```

```
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table orders(
```

```
-> order_id int primary key,
```

```
-> customer_name varchar(50),
```

```
-> product)id int,
```

```
-> quantity int,
```

```
-> ^C
```

```
mysql> create table orders(
```

```
-> order_id int primary key,
```

```
-> customer_name varchar(50),
```

```
-> product_id int,
```

```
-> quantity int,
```

```
-> foreign key(product_id)
```

```
-> );
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near ')' at
line 7
```

```
mysql> show tables;
```

```

+-----+
| Tables_in_cg_mysql |
+-----+
| company_staff      |
| employees          |
| products           |
+-----+
3 rows in set (0.00 sec)

```

```

mysql> drop table products;
Query OK, 0 rows affected (0.02 sec)

```

```

mysql> show tables;
+-----+
| Tables_in_cg_mysql |
+-----+
| company_staff      |
| employees          |
+-----+
2 rows in set (0.00 sec)

```

```

mysql> create table departments(
    -> DID int primary key,
    -> DNAME varchar(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

```

```

mysql> create table Doctors(
    -> Doc_id int primary key,
    -> salary decimal(10,2),
    -> DID int,
    -> ^C
mysql> create table Doctors(
    -> Doc_id int primary key,
    -> Doc_name varchar(50),
    -> salary decimal(10,2),
    -> DID int,
    -> foreign key (DID),
    -> references departments(DID)
    -> on delete set null on update set null
    -> );

```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'references departments(DID) on delete set null on update set null)' at line 6

```

mysql> create table Doctors(
    -> Doc_id int primary key,
    -> Doc_name varchar(50),
    -> salary decimal(10,2),

```

```

-> DID int,
-> foreign key (DID),
-> ^C
mysql> create table Doctors(
-> Doc_name varchar(50),
-> ^C
mysql> create table Doctors(
-> Doc_id int primary key,
-> Doc_name varchar(50),
-> salary decimal(10,2),
-> DID int,
-> foreign key (DID)
-> references departments(DID)
-> on delete set null
-> on update set null
-> );
Query OK, 0 rows affected (0.07 sec)

```

```

mysql> desc departments;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DID   | int           | NO   | PRI | NULL    |       |
| DNAME | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

```

mysql> insert into departments values(
111, 'Cardiologist'), (222, 'Dermatologist'), (333, 'Neurologist'), (444, 'Orthopedic
Surgeon');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

```

```

mysql> select * from departments;
+-----+-----+
| DID | DNAME          |
+-----+-----+
| 111 | Cardiologist   |
| 222 | Dermatologist  |
| 333 | Neurologist    |
| 444 | Orthopedic Surgeon |
+-----+-----+
4 rows in set (0.00 sec)

```

```

mysql> desc doctors;
+-----+-----+-----+-----+-----+-----+
| Field   | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Doc_id  | int           | NO   | PRI | NULL    |       |
| Doc_name | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

```

salary	decimal(10,2)	YES		NULL		
DID	int	YES	MUL	NULL		
+-----+-----+-----+-----+-----+-----+						

4 rows in set (0.00 sec)

```
mysql> insert into doctors values( 1, 'arjun reddy', 50000.0,222),(2,'tony stark',60000.0,333),(3,'victor',90000.0,444),(4,'vijay',50000.0,111);
ERROR 1136 (21S01): Column count doesn't match value count at row 3
```

```
mysql> INSERT INTO doctors VALUES
-> (1, 'arjun reddy', 50000.0, 222),
-> (2, 'tony stark', 60000.0, 333),
-> (3, 'victor', 90000.0, 444),
-> (4, 'vijay', 50000.0, 111);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select * from doctors;
+-----+-----+-----+-----+
| Doc_id | Doc_name | salary | DID |
+-----+-----+-----+-----+
| 1 | arjun reddy | 50000.00 | 222 |
| 2 | tony stark | 60000.00 | 333 |
| 3 | victor | 90000.00 | 444 |
| 4 | vijay | 50000.00 | 111 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> delete table departments DID=111 where DID=111;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'table departments DID=111 where DID=111' at line 1
mysql> delete departments DID=111 where DID=111;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'DID=111 where DID=111' at line 1
mysql> delete from departments where DID=111;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from doctors;
+-----+-----+-----+-----+
| Doc_id | Doc_name | salary | DID |
+-----+-----+-----+-----+
| 1 | arjun reddy | 50000.00 | 222 |
| 2 | tony stark | 60000.00 | 333 |
| 3 | victor | 90000.00 | 444 |
| 4 | vijay | 50000.00 | NULL |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> update departments set DID=555 where DID=null;
```

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> select * from doctors;

Doc_id	Doc_name	salary	DID
1	arjun reddy	50000.00	222
2	tony stark	60000.00	333
3	victor	90000.00	444
4	vijay	50000.00	NULL

4 rows in set (0.00 sec)

mysql> select * from departments;

DID	DNAME
222	Dermatologist
333	Neurologist
444	Orthopedic Surgeon

3 rows in set (0.00 sec)

mysql> update departments set DID=666 where DID=222;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from departments;

DID	DNAME
333	Neurologist
444	Orthopedic Surgeon
666	Dermatologist

3 rows in set (0.00 sec)

mysql> select * from doctors;

Doc_id	Doc_name	salary	DID
1	arjun reddy	50000.00	NULL
2	tony stark	60000.00	333
3	victor	90000.00	444
4	vijay	50000.00	NULL

4 rows in set (0.00 sec)

mysql> notee